

MEDICALAI · HARRI JUNTUNEN · POC

Clock, ledger, and enclosure for the 72-hour lone casualty

A Proof-of-Concept briefing on the MedPod hardware baseline and the companion app — the physical hold and the temporal hold for a conscious casualty alone for three days — aligned with the PCC doctrinal window. Exploratory build; distribution / App Store / MDR acceptance deferred.

PREPARED FOR Harri Juntunen / MedicalAI	STATUS Proof of Concept (PoC)	DATE 6 Sep 2026 · 72 h hold	SCOPE Exploratory — not productisation
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One scenario. Hold rescoped to 72 hours.

72 h

Hold from point of injury — conscious, self-treating, alone; occasional buddy contact. Inside the ~72 h PCC working ceiling.

0 net

No radio, cellular, or satellite in the operating state. Provisioning networked; ops are not.

UGV

No aeromedical. Recovery by unmanned ground vehicle or one-person vehicle + CONTAI AutoLoader

- **Shelter:** CONTAI MedPOD™ enclosure — Deliverable 2
- **Doctrine:** TCCC 01 May 2026 · JTS PCC CPG ID 91 · AJP-4.2 Ed A V1
- **User training:** TCCC All Service Member — 6–8 hours, at least 6 months stale

Why it matters: Doctrine assumes a caregiver arrives; the PoC tests whether enclosure + clock can hold a conscious solo casualty inside the ~72 h PCC window without claiming productisation.

Hold sits inside the ~72-hour PCC window. The gap is solo, not duration.

Nearly every intervention that improves survival in prolonged care either requires a second person — or becomes actively dangerous performed alone.

DOCTRINAL FIT

72 hours

- Matches the JTS PCC working ceiling
- Antibiotic prophylaxis ≤ 72 h ends with the hold
- Prior art: Role 2 ~48 h — stretch is ~1.5×, not weeks

STILL UNCOVERED

Solo care

- Doctrine assumes a caregiver for critical acts
- Self-applied tourniquet failure without outcome checks
- Only 14% arrive Role II/III with prehospital care documented

Why it matters: Stretch vs Role 2 (~1.5×) is modest; the uncovered gap is solo care — second-person acts and undocumented prehospital care (D1).

Two deliverables in scope. One as context.

D1 · Context Lone Casualty Knowledge Model Clinical domains, KR-01 thesis, uncovered hours, and the interventions that fail alone. Source of the product thesis — not briefed here. CONTEXT ONLY	D2 · Hardware MedPod Hardware Baseline Ventilated pressure-management enclosure. Insulation, not chemical heat. Fail-open life support. Soft BOM ~USD 552 / protected 672+; powered tier optional. THIS BRIEFING	D3 · Software MVP iPhone Application Clock and ledger before teacher. Five capabilities. Calculation firewall. iPhone 17 Pro Max · 12 GB · on-device only. THIS BRIEFING
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The intuitive capsule is wrong in three quantified ways.

INVERSION 1 · SEALING

~34 min

to 5% CO₂ in a sealed pod

About an hour to unconsciousness. Ventilation is the primary life-safety system — fail-open, zero power. Nothing else kills this fast.

INVERSION 2 · HEAT

0 W

active heating to ≈ -10 °C

Metabolic heat + 53 mm foam over the body and 85 mm beneath closes the balance. Chemical heat was always a 10-hour answer to a 72-hour problem (~7× short).

INVERSION 3 · REAL THREAT

42–48 mmHg

pelvis / heels on bare litter

Pressure injury, not cold, dominates. Stage 3–4 forms well inside a week. The best insulator — a vacuum mattress — is among the worst support surfaces.

Why it matters: Product order flips: ventilation integrity and pressure management first; warming almost free once insulation is right (D2 three inversions).

A sealed pod is a coffin on a one-hour timer.

- **0.22 L/min CO₂** from a resting 80 kg adult into ~150–400 L free air — occupational 0.5% in ~7 min; 5% intoxication in ~34 min; LOC at ~10% / ~68 min
- **Required ventilation:** 0.5–2 L/s per casualty, fail-open, no closeable valve — structurally impossible to seal, not instrumented
- **Face at hooded aperture** so exhaled CO₂ leaves without mixing; pod then needs only trickle ventilation
- **Counterflow recuperator** (~70% effective) drops the 12.6 W trickle penalty to ~3.8 W; HME recovers ~11.5 W and 287 g water/day
- **Excluded:** combustion / catalytic heaters inside the volume; air-activated chemical warmers add ~36% to O₂ demand

0.2%

of day 1 occupied by the entire sealed-pod timeline

4,320

minutes in the 72-hour mission

54 L/min

naive vent at -10 °C costs ~40 W — over half metabolic budget

24%

of heat budget is respiration — HME is best \$/g intervention

Why it matters: CO₂ kills in ~minutes when sealed; power must never be required to breathe — fail-open vents are life safety, not a comfort feature (D2).

Three coupled constraints — how the design answers.

1 · CO₂ / VENTILATION

Seal vs breathe

Problem: sealed ~150 L → ~34 min to 5% CO₂, ~1 h LOC; fastest killer.

Design: fail-open vents (no closeable valve); face at hooded aperture; trickle 0.5–2 L/s; counterflow recuperator (~70%) so vent doesn't steal heat; HME recovers respiratory heat/water.

Hard rule: power must never be required to breathe; powered tier still fail-open.

2 · HEATING NEED

72 h vs chemical heat

Problem: HPMK chemical heat ~10 h vs 72 h (~7× short); air-activated warmers raise O₂ demand; combustion/catalytic inside = CO (no exit).

Design: insulation first — ~53 mm foam over / ~85 mm under closes balance on ~70 W metabolism to ≈ -10 °C with **0 W active heat**; electrical heat only supplemental (powered tier) below ≈ -10 °C or collapsed metabolism, energy preferably outside occupied volume.

Thesis: active heat is last resort — never the product.

3 · THERMAL SIGNATURE

Warmth vs concealment

Problem: hypothermia doctrine wants warmth; concealment hates a hot blob; Mylar-outward / aluminium-out creates a **cold rectangle** on warm ground — often more conspicuous.

Design: same insulation grams buy survival **and** smaller ΔT to ambient; radiant barrier **metal face inward** across air gap (never outward as camouflage); matte outer shell; no reflective sunshade as primary cold-signature strategy; screen-off / deep-red UI for phone (KR-04).

Buy: survival + smaller ΔT with the same grams.

Bottom line: Insulation settles the heat–signature conflict; fail-open vent settles the seal–life conflict; active heat is last resort, never the product thesis.

Insulation, not chemical heat. Pressure injury dominates.

THERMAL

~70 W metabolic · 24 °C interior

- -10 °C: 53 mm over body / 85 mm beneath (XLPE-eq.)
- -25 °C: ~80 mm / 128 mm — Arctic is a thickness decision
- Base needs ~1.6× canopy R-value; ground is the sink
- Blizzard bag 7.5 TOG ≈ 26 mm — ~⅓ of -10 °C need
- Signature free: outer ΔT only 1.3–3.2 K; never metallise outward

PRESSURE

Vacuum mattress tension

- Bare litter: pelvis ~42 mmHg, heels ~48 mmHg
- “32 mmHg is safe” is a documented myth
- No published unattended multi-day pressure solution (still critical inside 72 h)
- Baseline: zoned foam + sacral/heel relief + mouth-inflatable tilt bladders (~USD 16, 300 g)
- Largest technical risk in the concept remains

Why it matters: Insulation closes the heat balance on ~70 W metabolism to ≈ -10 °C; chemical heat was a 10 h answer. Pressure injury (not cold) dominates inside 72 h — tilt bladders make nursing possible alone (D1 KR-08 / D2).

Mass stopped being scarce. Length binds instead.

\$552

Soft-pod material BOM · protected 672+ ·
powered tier separate

47.3 kg

Protected mass (soft + frame + aramid) ·
~110 kg ground allowance

~10 s

CONTAI AutoLoader claim — pickups, vans,
ATV, snowmobile

CONSTRAINT	OLD (AEROMEDICAL)	NEW (GROUND)	DESIGN IMPLICATION
Mass budget for pod	~30 kg	~110 kg	Insulation, loader frame, armour option all affordable
Binding dimension	UH-60 rack width	Pickup bed 1700 mm	Target ~2000 × 600 mm; modest overhang OK
Armour / interface	Ruled out by mass	Reopened	High-level option; AutoLoader geometry still unpublished

Powered tier exists (large battery + compute/display/heat) on top of the fail-open passive baseline — see BOM / Kit / Powered slides for BOM, kit, and powered affordances. Material prices indicative list only; no labour/tooling/margin.

MedPod BOM — soft pod + protected.

Material cost only · indicative list prices · no labour, tooling, or margin · D2-verified line items

ITEM	QTY	USD	KG	FUNCTION
XLPE foam base, 100 mm zoned	1.5 m²	194	7.20	R≈2.7 + sacral/heel relief
XLPE foam canopy, 50 mm	2.0 m²	129	4.80	R≈1.4
TPU-coated nylon 6.6 ripstop	6.0 m²	70	0.72	Shell, RF-weldable
Aluminised PET 12 µm	2.0 m²	6	0.03	Radiant barrier, metal face inward
Counterflow recuperator	1	10	0.25	~70% recovery, no power
HME respiratory	3	9	0.02	+11.5 W, water recovery
Mouth-inflatable tilt bladders	2	16	0.30	Self-offload
Drag handles/hardware	6 pt	45	0.90	One-person + UGV
Zips/closures FR	1 set	35	0.55	Full-length + limb ports
Graduated urine interface	1	12	0.20	Master field monitor
Vent screens fail-open	2	8	0.05	Cannot be closed
Reflective sunshade detachable	1	18	0.35	Hot case
Subtotal soft pod	—	552	15.37	
AutoLoader frame	1	120	9.00	Rails, pick points
Fragment aramid 3.9 m²	—	unpriced	22.90	Detachable
Protected total	—	672+	47.27	~60 kg ground recovery

Optional tiers: vacuum mattress USD 749 / 5.4 kg; alternating-pressure overlay+pump USD 400 / 2.5 kg (recommend). **Reject** aerogel and VIP — cost/fragility vs XLPE without commensurate field gain.

Why it matters: Each line buys a physics job — foam for R-value + sacral relief, recuperator/HME for vent heat, fail-open vents for CO₂, urine interface as master field monitor, frame for AutoLoader, detachable frag for FPV threat (D2).

Kit — medicines and instruments (72 h hold).

Separate from structural BOM · Fixed-dose cards only · Cite TCCC / JTS · Surface conflicts; do not resolve

MEDICINES · 72 H / 3 DAYS

Fixed-dose only

- **Cefadroxil 1 g PO** once daily × 3 — TCCC 01 May 2026. Conflict: JTS ID 24 still names moxifloxacin — surface both; do not silently pick.
- **Analgesia:** pre-portioned daily pouches × 3 (physical constraint, not a software ceiling). TCCC CWMP path; no opioids invented here.
- **Enoxaparin 30 mg BD** once haemostasis achieved and >48 h expected — JTS CPG ID 36; only if criteria met.
- TXA / other controlled acute resus drugs omitted from this kit page — not listed as MedPod hold field-kit line items in D1.

INSTRUMENTS / CONSUMABLES

Self-care + handover

- Combat-application tourniquet × ≥1 (limb)
- Haemostatic / packing gauze
- Splint (distal pulse checks)
- External thermometer (iPhone has none — kit line item)
- Graduated urine measure (also structural BOM interface)
- Fitbit or equiv BLE HR — MedPod default monitoring; trend only; SpO₂ not clinical
- Spare HME cartridges
- Oral care kit (KR-08 nursing/morale)
- Buddy-contact / handover print or QR-capable display path

Not a Role 2 pharmacy. PoC ships fixed-dose protocol cards only — no dose calculation in the app. Fixed doses + calculation firewall are **safety architecture for PoC**, not store-compliance theatre. Content is versioned and attributed; conflicts stay visible.

Why it matters: Cefadroxil vs moxifloxacin conflict must stay visible (KR-09). Tourniquet outcome gates catch confident skippers. Thermometer is kit because iPhone has none; Fitbit is trend-only — SpO₂ never clinical (D1/D3).

Powered MedPod — large battery on a fail-open baseline.

Programme decision: MedPod **may** carry a large battery, so electricity can be available. This is a powered tier on top of the passive envelope — not a replacement for zero-power life safety.

HARD RULES

— Ventilation remains fail-open / zero-power

— power must not be required to breathe

— No combustion / catalytic heater in the occupied volume (CO)

— Electrical heat is supplemental (below ~-10 °C or collapsed metabolism); stored energy outside occupied volume preferred

PROGRAMME NOTE

— Lab PC path vs field phone — shared ledger schema

— Powered tier changes BOM, mass, and emissions / signature profile

— Costs below labelled TBD where unknown — no invented prices

POWERED AFFORDANCE	ROLE	USD	NOTE
Large Li battery + BMS + charge port	Pod-side power	TBD	Sizes Wh vs heat + compute + charge budget
Gaming PC or equivalent	Lab/MVP local LLM + psych layer	TBD	Lab path; not assumed in field phone ops
Interior display	Deep-red / low-luminance UI	TBD	Signature discipline inside pod
Outward-facing cameras	SA / UGV approach	TBD	Emissions/signature trade — field SOP
Resistive heating pads	Supplemental heat only	TBD	Below ~-10 °C or collapsed metabolism
Alternating-pressure micro-pump	Pressure overlay	~400	Already optional overlay+pump tier
Fitbit BLE receiver / USB dongle	Default trend sensor path	TBD	PC lab; phone uses native BLE
LED task light (deep-red)	Task lighting	TBD	Signature discipline
USB charge for phone / Fitbit	72 h endurance	TBD	Across hold; battery budget item

Why it matters: Battery enables PC, display, cameras, supplemental heat — still on a fail-open vent baseline. Lab cameras ≈ RPM bedside awareness; field emissions are a signature trade (KB RPM theme / D2).

A clock and a ledger before it is a teacher.

C1 The Clock

Tourniquet 2 h / 6 h gates, drug intervals, q1–2 h repositioning, q6 h splint checks, dressing age. The irreplaceable function.

C2 The Ledger

Append-only clinical record — interventions, doses, measurements — timestamps from the device, not from failing memory.

C3 Outcome-gated guidance

Closed-set procedural cards gated on observable outcomes (e.g. distal pulse), not step confirmation.

C4 Self-measurement

Pulse count, urine volume, radial pulse Y/N, timed cognitive task vs personal baseline — four that survive self-admin.

C6 Psychological continuity

Nursing-as-morale, extraction log without shame (KR-07/08), timed cognitive vs baseline, procedural presence checks — continuity, not therapy.

C5 The Handover — DD Form 1380 / MIST

Packet on demand and on dead-man trigger. Readable by a medic with no app and no network — screen and QR. Strongest product argument; cheap to build.

Why it matters: Trustable clock/ledger before “smart” AI — evidence and trust gaps argue measure-first PoC; agents that follow up when no medic is present map to prompts, not replacement of care (KB / D3 KR-01).

What breaks the obvious design.

CONSTRAINT	FIGURE / FINDING	ARCHITECTURAL CONSEQUENCE
Calculation firewall	Model must not compute	All arithmetic & decision rules in deterministic Swift; model does language only
No dose calc in PoC	Deferred (post-PoC)	Fixed-dose cards + Swift firewall — safety architecture now; App Store 1.4.2 / publisher path later
Speech / on-device	SpeechAnalyzer + SpeechTranscriber	VAD-gated; custom medical lexicon; no SFSpeechRecognizer
Target device	iPhone 17 Pro Max OR gaming PC / equiv (lab MVP)	Phone: field form-factor + memory tier; PC: heavier local model + Web Bluetooth lab path
Fitbit / wearable	Default MedPod monitoring sensor	Consumer BLE HR trends; SpO ₂ never clinical; ≤60 s sessions; signal loss is a finding
Operating floor	0 °C	Phone lives inside MedPod envelope (pod holds ~24 °C to ≈ −10 °C ambient)
No network in ops	NFR-2	Build-time link check + runtime assertion; zero calls in operating state

Pod holds the body. App holds the timeline.

DELIVERABLE 2

MedPod

- Ventilation integrity first
- Passive thermal to ≈ -10 °C
- Pressure management attempt
- Thermal envelope for the phone
- UGV / AutoLoader interface



SHARED
ASSUMPTIONS

DELIVERABLE 3

MVP App

- Clock across 72 hours
- Ledger that outlives memory
- Psychological continuity artefacts
- Handover without a network
- Phone field or PC lab MVP

- **Neither replaces doctrine** — both cite TCCC / JTS; content is versioned and attributed
- **Both assume no network in ops** — enclosure physics and app architecture share the isolation premise
- **Dependency is mutual:** phone needs the pod's 0 °C floor fix; pod needs the app's pressure / drug / dressing clocks

Continuity under failed extraction — not therapy.

KR-07: extraction is recurring, not terminal. KR-08: the nursing schedule is morale architecture. Gap: no CPG for suicide / psych deterioration — we refuse to invent one.

01 Nursing-as-morale

Reposition, ankle pumps, oral care as one calm artefact (KR-08).

02 Extraction log

Recurring events; failed attempt does not reset clock or shame (KR-07).

03 Cognitive micro-task

Timed vs personal baseline; score deltas only — no diagnosis.

04 Presence checks

Procedural prompts; fixed spoken strings — never open chat therapy.

05 Missed-check signal

Overdue checks remain a soft deterioration indicator.

06 Night / dark UI

Screen-off rest; deep-red; no bright psych animations.

— **We refuse:** suicide-risk scoring claims · AI-companion romance · unsupervised psych diagnosis · crisis UI that pretends to treat

Why it matters: KR-07/08 make nursing and extraction continuity the morale layer — not therapy. No CPG for psych deterioration; PoC measures continuity artefacts, not clinical efficacy.

Fitbit default. Four self-measures still load-bearing. PC lab vs phone field.

LAYER	MVP CHOICE	CONSTRAINT
Default sensor	Fitbit / equiv BLE HR	Trend only; SpO ₂ never clinical logic; loss of signal is a finding
Self-measures	Pulse · urine · radial Y/N · cognitive	Four that survive self-admin — wearable does not replace them
PC / lab MVP	Gaming PC / equiv	Web Bluetooth or USB dongle; heavier local LLM; shared ledger schema
Phone field	iPhone 17 Pro Max	Scheduled ≤60 s foreground; emissions / signature discipline applies here
Hold	72 hours	Clock, prophylaxis window, and presentation figures aligned

Why it matters: Screen light is visible 0.5–2 km ground / 1–6 km airborne — screen-off rest is clinical for field phone; lab PC may differ (D1 KR-04).

Survivability — what we claim and what we refuse.

No validated “X% more survive with MedPod + app.” Claiming one would be marketing. Gain = avoided specific failure modes.

WILL SAY (POC) Scoped mechanisms only — Applies to casualties who already cleared the acute filter (conscious, haemorrhage controlled enough to self-treat) — Targets post-acute killers doctrine ignores for <i>solo</i> care: asphyxia-from-sealing, hypothermia from thin bags / 10 h chemical heat, missed nursing clocks, self-aid TQ errors, missing handover (only 14% documented) — Pod+app can improve odds on those mechanisms; measure proxies in PoC	WILL NOT SAY No blanket % · no Role 2 replace — A single survivability % for all battlefield wounded — Fixes truncal/junctional haemorrhage (~majority of potentially survivable bleeds) — Eliminates pressure injury (largest residual risk) — Replaces Role 2 / surgery; or treats psych/suicide (no CPG)
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~2.6% · extremity HE

Of all battlefield deaths ≈ extremity haemorrhage a lone casualty might self-treat (Eastridge stack · D1)

~63–65% · self-TQ

Lower-limb self-TQ success; common error = skip distal pulse (D1)

~1 in 5 · UGV

UGV retrieval success; extraction recurring (KR-07)

~10%→~30%

Evac stretch mortality is about time to surgery — not this product

PoC measure: hypothermia on recovery · Stage 2+ sores by 72 h · TQ verified occlusion · prophylaxis completed · handover completeness — then *model* a delta, don't invent one.

Why it matters: Evidence gap (KB: AI health tools under-evaluated) — PoC measures, does not claim clinical efficacy.

PoC learning goals — measure before claiming.

SAFE

Model must not unsupervised-decide

Safety architecture, not store policy: human confirmation on every clinically consequential action; language proposes, Swift commits. Unsupervised clinical AI is refused in PoC.

BENCH

Ventilation & thermal bench

Prove fail-open CO₂ path and insulation balance with measurements — every MedPod number is still a calculation until bench. Prior art max ~48 h Role 2; HPMK heat 10 h; 72 h is stretch, not fantasy.

UX

Clock + ledger + Fitbit session + psych continuity

Prove trustable timers/ledger before “smart” features; Fitbit ≤60 s trend sessions; six psych artefacts without therapy claims. Evidence gap: PoC must measure, not claim efficacy.

DEFER

Post-PoC only

App Store / publisher / EU MDR class · powered-tier Wh sizing · AutoLoader geometry · distribution. Fixed-dose firewall stays; store gate is not the PoC hardest problem.

Prototype experiments next

- 01 Bench ventilation (fail-open CO₂ path) and thermal balance vs D2 calculations — measure, do not claim.
- 02 Clock + ledger UX on phone and/or lab PC — prove timers and append-only record before teacher features.
- 03 Fitbit (or equiv) ≤60 s BLE trend session — lab Web Bluetooth path; SpO₂ never clinical.
- 04 Psychological continuity artefacts (KR-07/08) — six items + hard refusals; continuity, not therapy.
- 05 Powered-tier smoke test: battery + display + optional cameras in lab; vent still fail-open; signature SOP for field later.
- 06 Pressure residual: tilt-bladder usability + optional alternating overlay; AutoLoader geometry when CONTAI data arrives.

MedicalAI · Harri Juntunen · PoC · Deliverables 2 (MedPod Hardware Baseline) & 3 (MVP Application Spec) · 6 September 2026 · Figures tagged for provenance in source documents; nothing yet bench-validated. App Store / MDR / distribution deferred.